

Skin autofluorescence, a non-invasive marker of advanced glycation end products: clinical relevance and limitations

Cidila Da Moura Smedo,¹ M'Balu Webb,^{1,2} Helen Waller,³ Kamlesh Khunti,^{1,2,3} Melanie Davies^{1,2,3}

¹The Leicester Diabetes Centre, University Hospitals of Leicester NHS Trust, Leicester General Hospital, Leicester, UK
²NIHR Leicester-Loughborough Diet, Lifestyle and Physical Activity Biomedical Research Unit, University Hospitals of Leicester, Leicester General Hospital, Leicester, UK
³Diabetes Research Centre, University of Leicester, Leicester General Hospital, Leicester, UK

Correspondence to

Dr M'Balu Webb, The Leicester Diabetes Centre, University Hospitals of Leicester NHS Trust, Leicester General Hospital, Gwendolen Road, Leicester, LE5 4PW, UK; balu.webb@uhl-tr.nhs.uk

Received 13 October 2016
 Revised 20 December 2016
 Accepted 8 January 2017

ABSTRACT

Advanced glycation end products (AGEs) are protein-bound compounds derived from glycaemic and oxidative stress that contain fluorescent properties, which can be non-invasively measured as skin autofluorescence (SAF) by the AGE Reader. SAF has been demonstrated to be a biomarker of cumulative skin AGEs and potentially may be a better predictor for the development of chronic complications and mortality in diabetes than glycated haemoglobin A1c. However, there are several confounding factors that should be assessed prior to its broader application: these include presence of other fluorescent compounds in the skin that might be measured (eg, fluorophores), skin pigmentation and use of skin creams. The aim of this article is to provide a theoretical background of this newly developed method, evaluate its clinical relevance and discuss the potential confounding factors that need further analysis.

INTRODUCTION

The significant prevalence of diabetes mellitus (DM) and associated long-term complications has emerged as a global public health crisis in the recent years. It is estimated that approximately 552 million people worldwide will be affected with diabetes by 2030, predominantly in low/middle-income countries.¹ Chronic hyperglycaemia is the main cause of diabetic complications and has adverse effects in different tissue and organ systems, which leads to a complex cascade of events that are not fully understood.² One of these events is the increased production of advanced glycation end products (AGEs) in tissues. These are a group of heterogeneous compounds produced by non-enzymatic attachment of glucose to proteins, lipids or nucleic acids through the Maillard reaction, which significantly alters their function by affecting molecular conformation, enhancing cross-linking, changing enzyme activity and reducing receptor recognition.² The accumulation of AGEs results in adverse structural and functional modifications of tissues as such AGEs are thought to play a central role in the development of cardiovascular (CV) and renal complications associated with diabetes.^{3–5}

Since specific skin AGEs (SAGEs) exhibit intrinsic fluorescence properties, they can be measured by skin autofluorescence (SAF) using an AGE Reader (AGE Reader, DiagnOptics, Groningen, the Netherlands), which is a simple and non-invasive technique.⁶ Validation studies have provided consistent evidence about the significant association of SAF with AGEs content in skin biopsies. In

particular, SAGEs content accounted for up to 76% of the variance in the SAF levels in a meta-analysis of three validation studies,^{6–8} indicating that SAF can act as a measurable variable of SAGEs accumulation. Furthermore, SAGEs have been demonstrated to be a more significant predictor of diabetic complications and mortality over time (5–10 years) than the short-term glycaemic memory reflected by haemoglobin A1c (HbA1c) (3–6 months).^{9–11}

Although SAGEs assessment has been revealed to be a valuable non-invasive tool to estimate future disease risk, it is important to identify SAF confounding factors prior to application in the clinical setting. The effects of other skin fluorophores, skin pigmentation and the use of emollients have been reported on the validity of measurements.^{12–14} This article provides a background on SAGEs formation, discuss the clinical relevance of this tool and critically review factors influencing the measurements and thus require further assessment.

FORMATION OF AGES

Chronic hyperglycaemia enhances the production and accumulation of AGEs endogenously and exogenously as a consequence of oxidative and non-oxidative reactions of amino acids and reducing sugars in the Maillard reaction (figure 1).¹⁵ Protein glycation is primarily initiated when the carbonyl groups of reducing sugars, such as glucose, interact non-enzymatically with the reactive amino (α -NH₂) groups of proteins (lysine or arginine residues). Consequently, this interaction forms an unstable aldimine compound, a Schiff base. This reaction is relatively fast and is highly reversible, depending on substrate concentrations and incubation time. The Schiff base can be rearranged producing a stable Amadori product (eg, HbA1c), which accumulates on both short-lived and long-lived proteins over a period of several weeks.¹⁶ The Amadori product undergoes oxidative degradation to generate highly reactive intermediate dicarbonyl compounds, known as α -oxoaldehydes (glyoxal, methylglyoxal, deoxyglucosones) (figure 1) that interact again with free amino groups of proteins. Then, complex rearrangements of condensation, dehydration, hydrolysis or cross-linking reactions occur so highly heterogeneous, often fluorescent, insoluble and irreversible group of AGEs are formed, which accumulate and damage long-lived proteins such as extracellular matrix collagens.^{15 16} The accumulation of most AGEs occurs by intracellular glycation,

To cite: Da Moura Smedo C, Webb M, Waller H, et al. *Postgrad Med J* Published Online First: [please include Day Month Year] doi:10.1136/postgradmedj-2016-134579

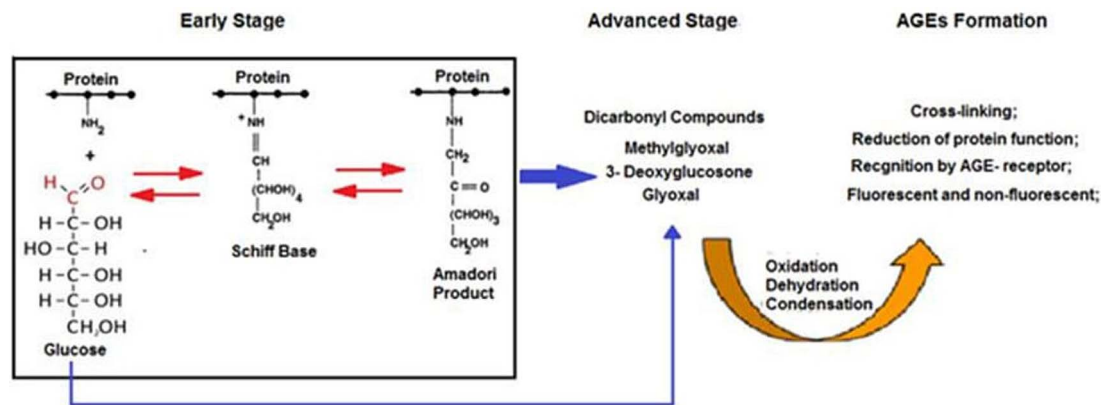
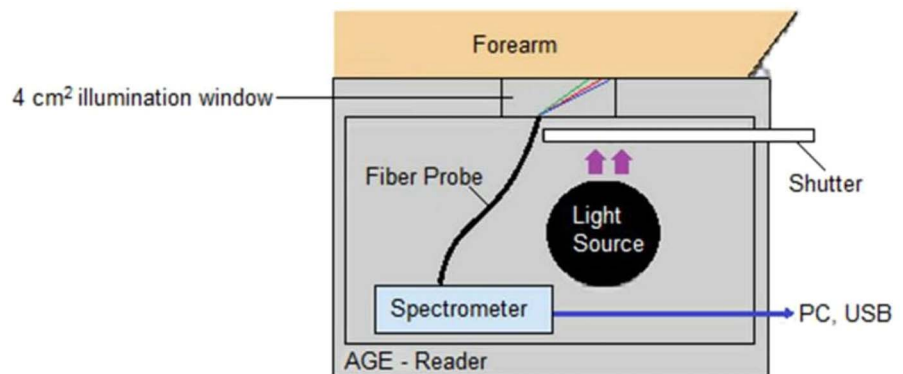


Figure 1 The Maillard reaction. Adapted from Nagai *et al.*¹⁷ This process involves three stages. In the early stage, glucose reacts with a free amino group of a protein to generate Schiff base that is subjected to further rearrangements to form Amadori product. These both processes are reversible. Then, reactive dicarbonyl compounds are formed at the advanced stage by either from the degradation of Amadori product or metal-catalysed autooxidation of glucose. Subsequently, irreversible advanced glycation end products (AGEs) are formed.

Figure 2 Schematic view of the advanced glycation end product (AGE) Reader. Adapted from Gerrits *et al.*²⁶ First, two calibration records are performed when the shutter is closed. One from white reflection standard and another from a dark current measurement. The light source will excite light when the shutter is opened to illuminate the patient's forearm skin through the illumination window. Subsequently, the emitted light from the skin is transferred through the fibre probe to the integrated spectrometer to transmit the data to a computer via USB connection and display the skin autofluorescence value on the screen.



cross-link formation in the extracellular matrix and interaction with specific cellular receptors.¹⁶

So far, a number of AGEs have been identified in the skin and these can be distinguished according to their fluorescence properties and cross-linking structures. These are fluorescent cross-link AGEs (pentosidine), non-fluorescent protein adducts (N-carboxymethyl-lysine (CML), N-carboxyethyl-lysine (CEL), glucosepane and fructose-lysine) and non-fluorescent protein cross-links (glyoxal-lysine dimer and methylglyoxal-lysine dimer).¹⁸

The degree of AGEs accumulation in the body is associated with their respective increased production, decreased degradation and renal clearance.^{19 20} The kidney is the predominant site of AGEs clearance, where filtered AGEs are degraded by renal proximal tubular cells and those remaining are excreted in the urine.²¹ However, when haemodynamic and metabolic changes occur due to exacerbated oxidative stress, the integral processes involved in the kidney function are interrupted.²² Consequently, it leads to renal impairment characterised by glomerular filtration damage and subsequent reduction in AGEs clearance, which may promote renal injury by further increasing body's tissue and circulating AGE pool.²³

THE PRINCIPLES BEHIND THE MEASUREMENT OF SAGES

Most AGEs that are evaluated *in vivo* or *in situ* can be detected in different specimen types and work as biomarkers of chronic

and age-related diseases.¹⁸ Several biochemical and immunochemical assays have been developed to determine levels of circulating AGEs. However, due to the heterogeneity of AGEs, cost, lack of reproducibility and high physiological fluctuations,¹² an alternative assessment of AGEs accumulation in the tissue is beneficial.

Since 2004, a simple, accurate and non-invasive screening technique SAF has been widely used in clinical trials to determine (in real time) tissue AGE accumulation in patients with diabetes, CV and renal complications: The AGE Reader (figure 2).⁶ SAF is directly correlated with both fluorescent (pentosidine) and non-fluorescent AGEs (CML and CEL) in the skin biopsies and so can be applied as an estimation method of SAGES accumulation.⁶

This device illuminates a subject's dominant lower forearm avoiding skin abnormalities. When the device is turned on, an ultraviolet (UV) tube emits UV light with peak intensity at 370 nm, which excites the AGEs in the skin that have characteristic autofluorescence properties in a frequency range of 300–420 nm. The autofluorescent light is measured with a spectrophotometer in the range of 300–600 nm. Normally, for each SAF value, three consecutive records are performed at three different sites of the same forearm, within a period of approximately 2 min. SAF is expressed in arbitrary units (AU) as the ratio between the emission light intensity in the 420–600 nm range and the excitation light intensity between 300 and

420 nm multiplied by 100 to compensate for the effect of skin pigmentation on autofluorescence by light absorption. Data from the spectrometer provide the SAF value which is displayed on the screen immediately, together with an age-corrected reference value.²⁴ The intra-individual and interday precision of this instrument has been tested previously in diabetic participants, giving a mean relative error of 4.2%–5%.²⁵

The skin reflectance percentage (SR%) is automatically expressed by the average intensities of the UV light reflected from the skin from a white reference standard in the 300–420 nm range. This calculation allows the reliable measurement of SAF independent of skin colour if the SR% is between 6% and 10%, which covers Fitzpatrick skin colour classes I–IV. However, when SR<6% (Fitzpatrick V–VI), no measurements are given by the AGE Reader.¹³ The Fitzpatrick scale (table 1) is a semiquantitative classification for assessing patient's skin colour by cutaneous response to UV.²⁷

CLINICAL RELEVANCE

Diabetes and chronic complications

There is consistent evidence that SAF is directly correlated with DM and development of diabetic complications (table 2). In a cross-sectional study, SAF values were considerably higher in patients with type 2 diabetes (T2DM) (n=973) than healthy subjects (n=231) across age groups.²⁸ The SAF values were increased in the presence of diabetes microvascular and macrovascular complications compared with those without complications (3.12 AU and 2.57 AU, respectively; p<0.001), independently of recent HbA1c levels. This agreed with similar SAF values reported from a prospective study in a multicentre secondary care setting.³

The predictive value of SAF has been proposed to be an important early surrogate marker for the harmful consequences of hyperglycaemia and oxidative stress over time and chronological ageing, where it may be superior to other risk predictors of diabetes (or chronic complications) such as HbA1c and diabetes duration. Accordingly, in a longitudinal study, SAF predicted the glycaemic and renal status of the preceding 10 years in elderly participants.²⁹ The development of microvascular complications was assessed by patients with T2DM (n=881) with good metabolic control of HbA1c 6.6%. After a mean follow-up period of 3 years, SAF at baseline revealed to be associated with the progress of microvascular complications,

predominantly for neuropathy and nephropathy but not for retinopathy.¹¹ These associations were also observed in a study performed in patients with type 1 DM (T1DM), where SAF levels were correlated with nephropathy and neuropathy but again not with retinopathy.³⁰ Nevertheless, SAF has demonstrated to be significantly correlated with diabetic neuropathy and retinopathy and with past long-term glycaemic control in a study of T1DM Japanese subjects over 15 years.³¹

CV risk and mortality

SAF is an important marker for the induction and development of vascular complications that can predict CV risk and death (table 2), when SAF values >2.0 AU over 5 years.¹⁰ CV risk is conventionally assessed according to age, blood pressure, lipid profile, smoking and DM;³² however, SAF measurements have shown additional value to the UK Prospective Diabetes Study (UKPDS) risk engine for CV risk assessment.³³ In this study, 25% of patients with a low UKPDS risk score (<10%) were reclassified at high risk to develop fatal and non-fatal CV events after the addition of SAF and vice versa. Moreover, SAF was able to predict those at high risk for fatal CV event in the next 10 years within the group with a high-risk score (>10%).³³

Recently, SAF values revealed to be elevated in patients with coexistent carotid artery disease (CAD) and peripheral artery disease (PAD) (3.28 AU) compared with subjects with only CAD (2.66 AU),³⁴ therefore indicating that SAF can potentially be a specific marker of atherosclerotic disease. Subjects with PAD showed higher SAF values than their control counterparts (2.77 AU vs 2.44 AU; p<0.001) independently of CV risk factors and comorbidity.³⁵ Moreover, patients with PAD (n=252) revealed that SAF was independently related with all-cause mortality and fatal or non-fatal major adverse CV events⁴ and predicted lower limb amputation³⁶ during a follow-up period of 5 years irrespective of DM and other CV risk factors.

CHALLENGES AND FUTURE PERSPECTIVES

Although SAGE accumulation appears to play a prominent role in characterising and predicting an individuals metabolic health, the AGE Reader contains several limitations that must be addressed to improve its relevance in the clinical setting.

Skin fluorophores

SAF assessment may not represent only the SAGEs content. The presence of endogenous fluorescent signals from skin fluorophores (eg, nicotinamide adenine dinucleotide) that are within the same excitation and emission ranges of the AGE Reader (350–410 and 420–600 nm, respectively) restricts the recognition of specific fluorophores on the total fluorescence signal.¹³ However, 76% of the variance in the SAF signal correlates with specific SAGEs content^{6–8} and the broad excitation peak of this device is ideal to assess the development of chronic complications in diabetes.³⁷ Therefore, indicating that less costly spectrometers or simple light sources can be implemented in this tool and so its application in the primary care can be extended.

Skin pigmentation

One important improvement for this instrument is to assess measurements in participants with darker skin pigmentation. The reliable analysis of SAGEs in darker skin subjects (SR<10%) has been particularly hindered, because strongly pigmented skin tends to absorb excitation light and so the identification of increased SAF values cannot be predicted.^{12 38}

Table 1 The Fitzpatrick scale according to skin pigmentation and cutaneous response to ultraviolet (UV)

Fitzpatrick class	Characteristics	Cutaneous response to UV
I	Bright white Northern European/British	Always burns and never tans
II	White European/Scandinavian	Burns easily and tans minimally
III	Fair Southern or Central European	Mild burn and average tanning ability to a light brown
IV	Light Brown Mediterranean, Asian or Latino	Minimal burn and tans easily to a moderate brown
V	Brown East Indian, Native American, Latino or African	Rarely burn and tans substantially
VI	Black African or Aboriginal ancestry	Almost never burns and tans abundantly

Adapted from D'Orazio et al.²⁷

Table 2 Predictive value of SAF to assess the development of diabetes complications and mortality

First author, year	Subjects (n)	SAF (AU)*	Reference group	Follow-up (years)	Outcome
De Vos, 2014 ⁴	PAD (252)	2.76	Per unit increase of SAF	5	Risk for all-cause mortality HR 2.01 and fatal or non-fatal MACE HR 1.82
Meerwaldt, 2007 ¹⁰	T1DM (48) T2DM (69) and C(43)	T1DM (0.016) T2DM (0.021) and C (0.010)	Per unit increase of SAF	5	CHD OR 7.9 and mortality OR 2.0
Gerrits, 2008 ¹¹	T2DM (881)	2.52	Per unit increase of SAF	3	Microvascular complications OR 2.02
Lutgers, 2009 ³³	T2DM (967)	2.69	Below or above median SAF	3/4	All-cause mortality HR 2.05 and CV events HR 1.46
De Vos, 2015 ³⁶	PAD (252)	2.76	Below or above median SAF	5	Amputation sHR 3.05

*SAF values at baseline.

AU, arbitrary units; CHD, coronary heart disease; CV, cardiovascular; MACE major adverse cardiac events; PAD, peripheral artery disease; SAF skin autofluorescence; sHR, subproportional HR; T1DM, type 1 diabetes mellitus; T2DM, type 2 diabetes mellitus.

Studies reported that this may be due to distinct absorption of light by skin compounds or scattering effects.^{12 39} The epidermis and dermis have UV radiation and visible region absorbing endogenous chromophores, melanin and haemoglobin, which absorb UVA light within the total reflected wavelength range between 320 and 420 nm.¹³ In particular, melanin has been shown to have a photoprotective role by acting as a physical barrier against UV radiation and as an absorbent chromophore that decreases the penetration of UV through the epidermis.⁴⁰ The total melanin content in the epidermis was demonstrated to be higher in darker skin subjects, while those with Asian and fairer skin differed by only twofold.⁴⁰ Indeed, SAF values between age-matched healthy Dutch and Chinese subjects revealed to be similar with SR>10% (p=0.253).^{38 41}

Currently, there is limited data about the SAF values in non-Caucasian patients with diabetes or diabetic complications. Most publications reporting other ethnic groups included East Asian subpopulations, predominantly from Japan or China.^{31 42 43} Therefore, it is essential to assess SAF levels in dark skinned subjects since they are the most prevalent skin type of the world and represent the group with highest incidence of diabetes.⁴⁴ Koetsier *et al*¹³ validated an adjusted algorithm to determine the UV light absorption in the spectral regions of melanin. Although these corrections have so far extended the range of SAF detection in darker skin subjects (up to Fitzpatrick class IV) with SR ≥6%, they were only developed in healthy subjects and have not been proved reliable in diabetes or diabetes complications from other ethnicities. Accordingly, a study involving a Saudi population (n=1999) revealed that the SAF risk score model may be not suitable for subjects with higher SAF values than those previously reported in Caucasian subjects,⁴⁵ because it underestimated subjects with low and high SR at high risk of developing CV disease.⁴⁵ This is therefore an urgent need for AGE Reader risk estimations to be tested in different ethnic populations.

Use of skin creams

SAF is strongly influenced by the application of different skin creams, leading to imprecise elevated SAF values and decreased SR%. This effect may be due to the creams' fluorescent characteristics from day cream or sunscreens that absorb UV radiation to protect the skin from adverse health effects. Day cream and sunscreen applied in the forearm provided 139% and 111% rise in the SAF measurements compared with 18% obtained by body lotion.¹⁴ Moreover, use of self-browning

dihydroxyacetone cream results in a chemical reaction with the stratum corneum to cause brown discolouration when applied to the skin and elevated absorption of UVA light, with a significant SAF increase of 298%.¹⁴

Washing the forearm with soap or using alcohol swabs does not reverse the effect of particular type of creams and can persist for many days. For example, sunscreen persisted for 4 days while self-browning cream 2 weeks,¹⁴ showing it is important to take these potential error sources into consideration to prevent discrepancies in the SAF measurements. The different prevalence of skin creams used between men and women may account for gender differences in SAGEs as previously reported.⁴⁶ This gender bias raises questions about the values obtained for CV and diabetes mortality between genders as estimated by the WHO country profile.⁴⁴

Thus far, the AGE Reader user manual suggests not conducting SAF measurements for at least 2 days after use of self-tanning agents on the forearm. These control measures have the potential to guarantee accuracy in the predictive values of SAF, which may be greater than those previously reported and where these precautions may not have been followed. Nevertheless, these measures do not necessarily detract from its application in the future. Currently, additional improvements in the AGE Reader software to limit significant fluctuations of SAF values are in development.¹⁴

CONCLUSION

AGE accumulation is demonstrated to play a pathophysiological role in the development of chronic complications in diabetes, including microvascular, macrovascular, CV complications and overall mortality. Most importantly, SAF assessment revealed to be an important biomarker of AGEs burden and a better predictor of diabetes complications because it represents a more 'long-term memory' of cumulative metabolic stress than HbA1c or other conventional risk factors (ie, smoking and blood pressure). This non-invasive tool is currently used for research purposes and whether it would be beneficial in daily clinical practice requires further studies, but its advantages constitute a consistent perspective of a variable suitable for monitoring interventions and limit the incidence of diabetes complications in the future. Thus far, SAF measurements contain several technical issues that are still being improved. Therefore, further research to expand the application of this technique is crucial as this is a practical, relatively inexpensive and non-invasive procedure available to be used in outpatient clinics.

Main messages

- ▶ The advanced glycation end product (AGE) Reader is a non-invasive, practical and mobile device with low educational or training input.
- ▶ Skin autofluorescence (SAF) assessment extends history of hyperglycaemia compared with conventional haemoglobin A1c.
- ▶ SAF is a measure of skin AGEs, which may be useful in large-scale studies and clinical setting, mostly when follow-up measurements are required to predict progression of diabetes mellitus-related complications and allow targeting of therapy
- ▶ SAF measurements are strongly influenced by the optical properties of the skin and so it is crucial to adjust measurements for these confounding factors to be able to compare results between subjects and different ethnicities.

Current research questions

- ▶ Does the accumulation of advanced glycation end products (AGEs) represent the pathology of diabetes or other pathological mechanisms?
- ▶ Does skin autofluorescence (SAF) also detects non-fluorescent AGEs as depicted by skin biopsies?
- ▶ What will be the reasonable time frame to assess improvements of the AGEs levels after treatment?
- ▶ Is SAF a suitable method to predict diabetic complications in non-Caucasian ethnicities?

Key references

- ▶ Meerwaldt R, Graaff R, Oomen P, *et al.* Simple non-invasive assessment of advanced glycation end product accumulation. *Diabetologia* 2004;47:1324–30.
- ▶ Gerrits E, Lutgers H, Kleefstra N, *et al.* Skin autofluorescence: a tool to identify type 2 diabetic patients at risk for developing microvascular complications. *Diabetes Care* 2008;31:517–21.
- ▶ Lutgers H, Gerrits E, Graaff R, *et al.* Skin autofluorescence provides additional information to the UK Prospective Diabetes Study (UKPDS) risk score for the estimation of cardiovascular prognosis in type 2 diabetes mellitus. *Diabetologia* 2009;52:789–97.
- ▶ Koetsier M, Nur E, Chunmao H, *et al.* Skin colour independent assessment of aging using skin autofluorescence. *Opt. Express* 2010;18:14416–29.
- ▶ Noordzij M, Lefrandt J, Graaff R, *et al.* Dermal factors influencing measurement of skin autofluorescence. *Diabetes Technol Ther* 2011;13:165–70.

Self assessment questions

Please answer true or false to the below statements.

1. The advanced glycation end product (AGE) Reader provides a rapid assessment of diabetes complications and cardiovascular risk prediction or mortality.
2. The reported skin autofluorescence (SAF) reference values in Chinese subjects is significantly higher than Caucasian subjects
3. Darker skin pigmentation can potentially affect SAF measurements
4. The effect of skin creams on SAF measurements may not fully be reversed by cleaning the forearm with water or alcohol swab.
5. The fluorescent characteristics detected by the AGE Reader represent only skin AGEs content.

Contributors CDMS wrote the manuscript and MW, HW, KK and MD reviewed and advised on each draft.

Funding This work was supported by the NIHR Leicester-Loughborough Diet, Lifestyle and Physical Activity Biomedical Research Unit which is a partnership between University Hospitals of Leicester NHS Trust, Loughborough University and the University of Leicester and the National Institute for Health Research Collaboration for Leadership in Applied Health Research and Care—East Midlands (NIHR CLAHRC—EM).

Competing interests None declared.

Provenance and peer review Not commissioned; externally peer reviewed.

REFERENCES

- 1 Whiting D, Guariguata L, Weil C, *et al.* IDF Diabetes Atlas: Global estimates of the prevalence of diabetes for 2011 and 2030. *Diabetes Res ClinPract* 2011;94:311–21.
- 2 Nowotny K, Jung T, Höhn A, *et al.* Advanced Glycation End Products and Oxidative Stress in Type 2 Diabetes Mellitus. *Biomolecules* 2015;5:194–222.
- 3 Noordzij MJ, Mulder DJ, Oomen PH, *et al.* Skin autofluorescence and risk of micro- and macrovascular complications in patients with Type 2 diabetes mellitus—a multi-centre study. *Diabet Med* 2012;29:1556–61.
- 4 de Vos L, Mulder D, Smit A, *et al.* Skin autofluorescence is associated with 5-year mortality and cardiovascular events in patients with peripheral artery disease. *ArteriosclerThrombVascBiol* 2014;34:933–8. doi: 10.1161/ATVBAHA.113.302731
- 5 Nongnuch A, Davenport A. Skin autofluorescence advanced glycosylation end products as an independent predictor of mortality in high flux haemodialysis and haemodialysis patients. *Nephrology (Carlton)* 2015;20:862–7.
- 6 Meerwaldt R, Graaff R, Oomen PH, *et al.* Simple non-invasive assessment of advanced glycationendproduct accumulation. *Diabetologia* 2004;47:1324–30.
- 7 Meerwaldt R, Hartog JW, Graaff R, *et al.* Skin autofluorescence, a measure of cumulative metabolic stress and advanced glycation end products, predicts mortality in hemodialysis patients. *J Am SocNephrol* 2005;16:3687–93.
- 8 Den Hollander NC, Mulder DJ, Graaff R, *et al.* Advanced glycation end products and the absence of premature atherosclerosis in glycogen storage disease Ia. *J Inherit Metab Dis* 2007;30:916–23.
- 9 Genuth S, Sun W, Cleary P, *et al.* Glycation and carboxymethyllysine levels in skin collagen predict the risk of future 10-year progression of diabetic retinopathy and nephropathy in the diabetes control and complications trial and epidemiology of diabetes interventions and complications participants with type 1 diabetes. *Diabetes* 2005;54:3103–11.
- 10 Meerwaldt R, Lutgers H, Links T, *et al.* Skin autofluorescence is a strong predictor of cardiac mortality in diabetes. *Diabetes Care* 2007;30:107–12.
- 11 Gerrits EG, Lutgers HL, Kleefstra N, *et al.* Skin Autofluorescence: A tool to identify type 2 diabetic patients at risk for developing microvascular complications. *Diabetes Care* 2008;31:517–21.

Review

- 12 Mulder D, Water T, Lutgers H, *et al.* Skin Autofluorescence, a Novel Marker for Glycemic and Oxidative Stress-Derived Advanced GlycationEndproducts: An Overview of Current Clinical Studies, Evidence, and Limitations. *Diabetes Technol Ther* 2006;8:523–35.
- 13 Koetsier M, Nur E, Chunmao H, *et al.* Skin color independent assessment of aging using skin autofluorescence. *Opt Express* 2010;18:14416–29.
- 14 Noordzij M, Lefrandt J, Graaff R, *et al.* Dermal Factors Influencing Measurement of Skin Autofluorescence. *Diabetes TechnolTher* 2011;13:165–70.
- 15 Tessier F. The Maillard reaction in the human body. The main discoveries and factors that affect glycation. *Pathol Bio* 2010;58:214–19.
- 16 Piarulli F, Sartore G, Lapolla A. Glyco-oxidation and cardiovascular complications in type 2 diabetes: a clinical update. *ActaDiabetol* 2013;50:101–10.
- 17 Nagai R, Jinno M, Ichihashi M, *et al.* Advanced glycation end products and their receptors as risk factors for aging. *Anti-Aging Medicine* 2012;9:108–13.
- 18 Gkogkolou P, Böhm M. Advanced glycation end products: key players in skin aging? *Dermatoendocrinol* 2012;4:259–70.
- 19 Weiss M, Erhard P, Kader-Attia F, *et al.* Mechanisms for the formation of glycoxidation products in end-stage renal disease. *Kidney Int* 2000;57:2571–85.
- 20 Uribarri J, Peppia M, Cai W, *et al.* Restriction of dietary glycotoxins reduces excessive advanced glycation end products in renal failure patients. *J Am Soc Nephrol* 2003;14:728–31.
- 21 Gugliucci A, Bendayan M. Renal fate of circulating advanced glycated end products (AGE): evidence for reabsorption and catabolism of AGE-peptides by renal proximal tubular cells. *Diabetologia* 1996;39:149–60.
- 22 Baynes JW, Thorpe SR. Glycoxidation and lipoxidation in atherogenesis. *Free Radic Biol Med* 2000;28:1708–16.
- 23 Kanwar Y, Wada J, Sun L, *et al.* Diabetic Nephropathy: Mechanisms of Renal Disease Progression. *Exp Biol Med (Maywood)* 2008;233:4–11.
- 24 Meerwaldt R, Links T, Graaff R, *et al.* Simple Noninvasive Measurement of Skin Autofluorescence. *Ann N Y AcadSci* 2005;1043:290–8.
- 25 Cleary P, Braffett B, Orchard T. Clinical and technical factors associated with skin intrinsic fluorescence in subjects with type 1 diabetes from the diabetes control and complications trial/epidemiology of diabetes interventions and complications study. *Diabetes Technol Ther* 2013;15:466–74.
- 26 Gerrits EG, Smit AJ, Bilo HJ. AGEs, autofluorescence and renal function. *Nephrol Dial Transplant* 2009;24:710–13.
- 27 D'Orazio J, Jarrett S, Amaro-Ortiz A, *et al.* UV Radiation and the Skin. *Int J Mol Sci* 2013;14:12222–48.
- 28 Lutgers HL, Graaff R, Links TP, *et al.* Skin autofluorescence as a noninvasive marker of vascular damage in patients with type 2 diabetes. *Diabetes Care* 2006;29:2654–9.
- 29 Rajaobelina K, Cougnard-Gregoire A, Delcourt C, *et al.* Autofluorescence of skin advanced glycation end products: marker of metabolic memory in elderly population. *J Gerontol A Biol Sci Med Sci* 2015;70:841–6.
- 30 Chabroux S, Canoui-Poitrine F, Reffet S, *et al.* Advanced glycation end products assessed by skin autofluorescence in type 1 diabetics are associated with nephropathy, but not retinopathy. *Diabetes Metab* 2010;36:152–7.
- 31 Sugisawa E, Miura J, Iwamoto Y, *et al.* Skin autofluorescence reflects integration of past long-term glycemic control in patients with type 1 diabetes. *Diabetes Care* 2013;36:2339–45.
- 32 Martín-Timón I, Sevillano-Collantes C, Segura-Galindo A, *et al.* Type 2 diabetes and cardiovascular disease: have all risk factors the same strength? *World J Diabetes* 2014;5:444–70.
- 33 Lutgers H, Gerrits E, Graaff R, *et al.* Skin autofluorescence provides additional information to the UK Prospective Diabetes Study (UKPDS) risk score for the estimation of cardiovascular prognosis in type 2 diabetes mellitus. *Diabetologia* 2009;52:789–97.
- 34 Noordzij MJ, Lefrandt JD, Loeffen EA, *et al.* Skin autofluorescence is increased in patients with carotid artery stenosis and peripheral artery disease. *Int J Cardiovasc Imaging* 2012;28:431–8.
- 35 de Vos L, Noordzij M, Mulder D, *et al.* Skin Autofluorescence as a Measure of Advanced Glycation End Products Deposition Is Elevated in Peripheral Artery Disease. *Arterioscler Thromb Vasc Bio* 2013;33:131–8.
- 36 de Vos L, Boersema J, Mulder D, *et al.* Skin autofluorescence as a measure of advanced glycation end products deposition predicts 5-year amputation in patients with peripheral artery disease. *Arterioscler Thromb Vasc Bio* 2015;35:1532–7.
- 37 Koetsier M, Lutgers H, Smit A, *et al.* Skin autofluorescence for the risk assessment of chronic complications in diabetes: a broad excitation range is sufficient. *Opt Express* 2009;17:509–19.
- 38 Koetsier M, Lutgers H, de Jonge C, *et al.* Reference Values of Skin Autofluorescence. *Diabetes TechnolTher* 2010;12:399–403.
- 39 Yudovsky D, Pilon L. Retrieving skin properties from in vivo spectral reflectance measurements. *J Biophotonics* 2011;4:305–14.
- 40 Brenner M, Hearing V. The protective role of melanin against UV damage in human skin. *Photochem Photobiol* 2008;84:539–49.
- 41 Yue X, Hu H, Koetsier M, *et al.* Reference values for the Chinese population of skin autofluorescence as a marker of advanced glycation end products accumulated in tissue. *Diabet Med* 2011;28:818–23.
- 42 Tanaka K, Tani Y, Asai J, *et al.* Skin autofluorescence is associated with severity of vascular complications in Japanese patients with Type 2 diabetes. *Diabet Med* 2012;29:492–500.
- 43 Liu C, Xu L, Gao H, *et al.* The association between skin autofluorescence and vascular complications in Chinese patients with diabetic foot ulcer: an observational study done in shanghai. *Int J Low Extrem Wounds* 2015;14:28–36.
- 44 World Health Organization. *Global status report on noncommunicable diseases*. Geneva: World Health Organisation, 2011.
- 45 Ahmad M, Damanhour Z, Kimhofer T, *et al.* A new gender-specific model for skin autofluorescence risk stratification. *Sci Rep* 2015;5:10198.
- 46 Mook-Kanamori M, Selim M, Takiddin A, *et al.* Ethnic and gender differences in advanced glycation end products measured by skin auto-fluorescence. *Dermatoendocrinol* 2013;5:325–30.

Answers

1. True
2. False
3. True
4. True
5. False



Skin autofluorescence, a non-invasive marker of advanced glycation end products: clinical relevance and limitations

Cidila Da Moura Semedo, M'Balu Webb, Helen Waller, Kamlesh Khunti and Melanie Davies

Postgrad Med J published online January 31, 2017

Updated information and services can be found at:

<http://pmj.bmj.com/content/early/2017/01/31/postgradmedj-2016-134579>

These include:

References

This article cites 45 articles, 10 of which you can access for free at:
<http://pmj.bmj.com/content/early/2017/01/31/postgradmedj-2016-134579#BIBL>

Email alerting service

Receive free email alerts when new articles cite this article. Sign up in the box at the top right corner of the online article.

Notes

To request permissions go to:

<http://group.bmj.com/group/rights-licensing/permissions>

To order reprints go to:

<http://journals.bmj.com/cgi/reprintform>

To subscribe to BMJ go to:

<http://group.bmj.com/subscribe/>